

CEN 354 Operating Systems

Spring 2020-2021

Instructor: Dr. Bariş ATA (bata@cu.edu.tr)

Office Hours: Wednesday, 13:00-16:00

Description:

This course introduces computer operating systems; processes, threads, interprocess communication, process scheduling, process synchronization, deadlocks, memory management and virtual memory, file systems - interface and implementation, mass-storage structure and management, input/output systems, examples from operating systems such as Linux and Windows.

Textbook:

- Operating System Concepts, Tenth Edition, Avi Silberschatz, Peter Baer Galvin, Greg Gagne, 2018, Wiley

References:

- Operating Systems Internals and Design Principles, William Stallings
- Operating Systems in Depth, Thomas W. Doeppner
- Modern Operating Systems, Andrew S. Tanenbaum, Herbert Bos

Course Schedule:

Week 1:	Introduction
Week 2:	Operating System Structures
Week 3:	Processes
Week 4:	Threads
Week 5:	Scheduling
Week 6:	Synchronization I
Week 7 :	Synchronization II
Week 8 :	Midterm
Week 9 :	Deadlocks
Week 10:	Memory Management
Week 11:	Virtual Memory
Week 12:	Mass Storage
Week 13:	I/O Systems
Week 14:	File Systems I
Week 15:	File Systems II

Grading: Assignments 30%, Midterm 10%, Final Exam 60%.